

Usability Analysis of Digital Signature Application Based on ISO 9241-11 Using Retrospective Think Aloud and User Experience Questionnaire

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Abstract—Applications for digital signatures are essential in supporting the efficiency and security of electronic administrative processes across various institutions, including research organizations. This research is intended to assess the usability of the DS BRIN Digital Signature application by employing ISO 9241-11 standards as the foundation for assessing effectiveness and efficiency. Additionally, a combination of the User Experience Questionnaire (UEQ) is utilized to evaluate satisfaction levels, while the Retrospective Think Aloud (RTA) method is applied to gather improvement suggestions from users' reflections on their experiences during interactions with the application. The findings reveal that while the application achieves a high effectiveness rate with a 93% task completion rate, its efficiency is notably low, with users reaching only 45% of optimal performance efficiency on average. This inefficiency is attributed to difficulties in navigation and non-intuitive features, where users often become disoriented or misuse the features. Additionally, results from the User Experience Questionnaire (UEQ) indicate low user satisfaction, with some aspects rated in "Bad" category. The study concludes that the DS BRIN application requires significant improvements in usability, especially in enhancing navigation and feature intuitiveness, to improve overall user experience and satisfaction.

Keywords—*usability analysis, user experience, retrospective think aloud, digital signature*

I. INTRODUCTION

In the process of electronic document authentication, digital signatures have become a widely adopted solution across various sectors, including the government sector. Compared to conventional signatures, digital signatures offer high efficiency and security [1]. This process ensures that documents remain unchanged after being signed and guarantees non-repudiation, meaning signers cannot deny having signed the document [2]. The adoption of digital signatures in government sectors is steadily increasing [3]. One application used in the government

environment in Indonesia is the Digital Signature BRIN (DS BRIN). However, the success of digital application implementation is measured not only by the functions or features provided but also by usability or ease of use [4]. With usability analysis, we can identify the actual needs and tasks of the user and provide clear, evidence-based recommendations for design improvements. This approach effectively reduces user adjustment time and errors while enhancing overall user productivity [5]. Several research have been carried out to evaluate the usability of digital signature applications through established methods. These studies often employ methods like the System Usability Scale (SUS) or the User Experience Questionnaire (UEQ) [6], [7]. Nonetheless, there seems to be a shortage of research utilizing a multi-method approach in this area up to now. Several methods employed to evaluate usability in this study include Usability Testing, UEQ, and Retrospective Think Aloud (RTA). The UEQ is a survey instrument that enables the measurement of various aspects of user experience. RTA is a qualitative method in which users are asked to reflect on their experience after using the application and articulate their thoughts during use. RTA facilitates a deeper collection of insights regarding user challenges or issues encountered during interaction with the application. Meanwhile, ISO 9241-11 provides a framework for evaluating user effectiveness, efficiency, and satisfaction in real-world contexts, allowing a comprehensive assessment of how the application meets user needs in practical environments [8]. This study aims to conduct a usability evaluation of the DS BRIN digital signature application using a combination of Usability Testing, UEQ and RTA based on ISO 9241-11. Through this multi-method approach, we intend to obtain a comprehensive insight into the user experience, evaluate the application's areas for improvement and suggest recommendations for future development.

II. SYSTEM OVERVIEW AND LITERATURE REVIEW

A. Digital Signature BRIN (DS BRIN)

BRIN has adopted the implementation of digital signatures through the Digital Signature BRIN (DS BRIN) system. DS BRIN offers several features, including signing documents, creating file categories, signing multiple documents, signature request from other users, and signing approval [9], as shown in Fig. 1. DS BRIN adopts cryptography-based technology, utilizing electronic certificates to sign and validate documents. The implemented cryptographic data transformation, will create a method to validate the authenticity of the source, ensure data integrity, and enable signer's non-repudiation [10], aiming to confirm the origin of the text and ascertain if any modifications have been made [11].



Fig. 1. DS BRIN Main Page with Description

DS BRIN has a total of 11 menus, with 8 menus dedicated to signing, and 3 menus for non-signing activities. Among the 8 signing menus, several menus have redundant functions, as shown in Fig. 2.

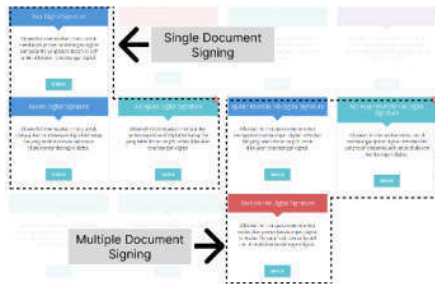


Fig. 2. Signing Menu Redundancy

B. ISO 9241-11

ISO 9241-11 is a part of the ISO 9241 international standard that focuses on the ergonomics of human-system interaction. The latest version, updated in 2018, emphasizes the definition and concepts of usability. This standard aims to guide designing, evaluating, and improving interactive systems to make them more user-friendly. ISO 9241-11 defines usability as: "The extent to which specified users can use a product to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use". This study uses ISO 9241-11 as a framework for measuring usability through usability testing, covering effectiveness, efficiency, and satisfaction.

C. User Experience Questionnaire

One direct method for measuring usability and user experience (UX) is through standardized questionnaires [12]. The User Experience Questionnaire (UEQ) is a tool that can assess data obtained from other evaluation methods by providing subjective evaluations of quality related to user experiences [13]. This ensures precision, authenticity, and supports users in expressing their feelings, opinions, and attitudes while engaging with a product [14]. The primary purpose of the UEQ is to facilitate a quick and straightforward assessment of the user experience of an interactive product, taking into account both practical and emotional quality [15]. The UEQ consists of six scales and encompassing 26 items, as illustrated in the Fig. 3.

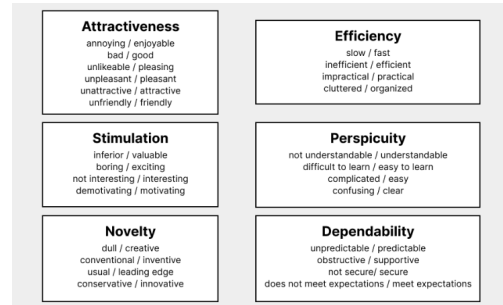


Fig. 3. UEQ Scales and Items

D. Retrospective Think Aloud (RTA)

Retrospective Think Aloud (RTA) is a method that can be used in usability testing [16]. In RTA, participants start by completing their tasks quietly and only express their thoughts afterward, using a video of their task execution [17]. RTA enables participants to focus on their current task, leading to a more organic interaction with the system. This approach also enables participants to recall and provide more meaningful reflections on their experience [18].

E. Related Works

Prior studies have focused on evaluating usability and user experience across digital platforms in diverse sectors to understand user interaction challenges and satisfaction levels. One study examined electronic signature usage in academic activities during the COVID-19 pandemic, employing the User Experience Questionnaire (UEQ) to assess pragmatic and hedonic qualities, both scoring highly, thereby supporting the system's positive reception among academic users [7]. In the commercial sector, a life insurance mobile application was evaluated using System Usability Scale (SUS) and ISO 9241-11, yielding insights into its effectiveness and efficiency but highlighting areas for interface improvement to aid non-tech-savvy users [19]. Another study on an IT support mobile app used SUS and RTA, revealing usability issues in navigation and suggesting interface updates to enhance technician performance and user satisfaction [20]. Collectively, these studies highlight the importance of structured usability assessments, forming a basis for this research's approach to evaluating the Digital Signature BRIN application with methodologies like UEQ, RTA, and ISO standards for a comprehensive user experience analysis.

III. RESEARCH METHODOLOGY

This study uses a mixed-methods approach to assess the usability of the DS BRIN Digital Signature application, using both quantitative and qualitative methods. The three methods utilized in this research are Usability Testing, RTA, and UEQ as shows in Fig. 4. Usability testing examines two of the three quality attributes outlined in ISO 9241-11's usability definition—effectiveness and efficiency. User satisfaction, the third attribute, is derived from the UEQ results. The RTA approach collects respondent insights, which serve as a basis for proposing design improvements. Each method offers a unique perspective on usability, together providing a thorough view of the quality of user interaction with the application.

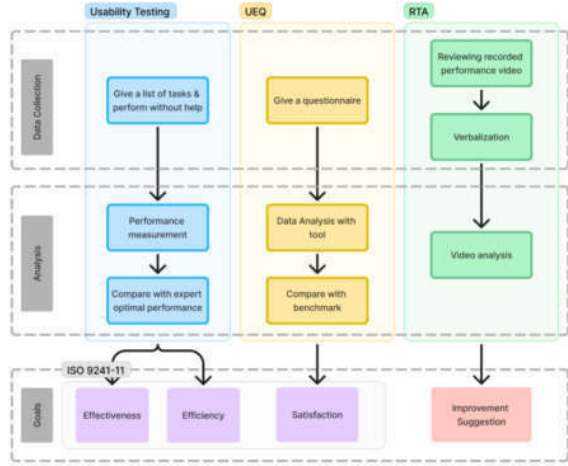


Fig. 4. Research Methodology

This study involved 10 beginner participants who were active users of the DS BRIN Digital Signature application and 5 expert users as a benchmark for assessing effectiveness and efficiency. Participants were selected based on the following criteria: (1) a minimum of three months' experience using the DS BRIN application; (2) a usage frequency of less than 20 times per month.

A. Usability Testing

Usability testing provides precise details on how people use computers and what issues they have with the tested concrete interface. This testing involves a group of users performing predetermined tasks while collecting data on time taken and errors made. In data collection, techniques and assistance such as mouse tracker applications and stopwatches can be used [21]. In this assessment, participants were instructed to finish a set of tasks within the DS BRIN application, based on predefined task scenarios. The testing instrument recorded all on-screen activities throughout the session. These recordings served as both analysis material for the researchers and as reference material for participants in conducting the Retrospective Think Aloud (RTA). This testing aimed to measure two of the three quality attributes within the ISO 9241-11 usability definition: effectiveness and efficiency. Effectiveness was evaluated based on the participants' success rate in completing the assigned tasks. Each successfully completed task was assigned a score of 1, while a score of 0 was given for tasks the participant failed

to complete. The effectiveness is determined using (1), while the efficiency is determined using (2) [22].

$$\bar{E} = \frac{\sum_{j=1}^R \sum_{i=1}^N n_{ij}}{NR} \times 100 \quad (1)$$

Notes:

$\bar{E}$ = Completion rate

R = Number of users

N = Number of tasks

n_{ij} = The result of task i by user j ; if the user successfully completes the task, then $n_{ij} = 1$, if not, then $n_{ij} = 0$.

The results from the effectiveness calculations were then compared with the optimal performance of expert users and statistically tested using Mann-Whitney U-Test to verify the significance of the effectiveness values, while efficiency was assessed based on the time and accuracy with which participants completed the tasks using Expert Relative Efficiency and then compares the test results with the ideal performance achieved by experts—defined in this research as users with extensive experience in using the application.

$$\bar{P}_E = \bar{E} \times \frac{R \sum_{i=1}^N t_{0i}}{\sum_{j=1}^R \sum_{i=1}^N t_{ij}} \quad (2)$$

Notes:

$\bar{P}_E$ = Expert Relative Efficiency

$\bar{E}$ = Beginners completion rate

R = Number of beginners users

N = Number of tasks

t_{0i} = The ideal time spent by expert to complete task i

t_{ij} = The time spent by beginner user j to complete task i . If the task is not successfully completed, then time is measured till the moment the user quits the task.

B. User Experience Questionnaire (UEQ)

The research used the UEQ method to assess respondents' satisfaction levels with the application. This measurement was conducted by calculating the mean score of all items within each scale category in the questionnaire. The calculations were performed using the UEQ Data Analysis Tool Version 12. The average score for each indicator was then compared with benchmark data available in the UEQ Analysis Tools. The benchmark dataset includes data from 21,175 individuals across 468 studies and is updated annually. Comparing the test analysis results with the benchmark data will illustrate the respondents' satisfaction level in using the tested application [23].

C. Retrospective Think Aloud (RTA)

To develop recommendations for improving the DS BRIN module, this study utilizes data gathered from the Retrospective Think Aloud (RTA) sessions. In these RTA sessions, respondents are asked to review video recordings of their usability testing sessions. Based on these recordings, the researcher prompts respondents with task-related questions to encourage them to verbalize any challenges they encountered while performing tasks, with this feedback being recorded. The data is then analyzed and summarized to identify features that require improvement, along with suggestions for enhancing those features.

TABLE 1. BEGINNER PARTICIPANTS TESTING RESULTS DATA

Respondent Code	Completion Time per Task (seconds)										Total Time	Completed Tasks
	Task											
	1	2	3	4	5	6	7	8	9	10		
R1	6	104	442	201	200	25	332	81	215	112	1718	7
R2	5	29	117	114	49	28	358	90	106	80	976	10
R3	5	8	69	93	50	19	183	51	32	19	529	10
R4	2	43	121	60	39	51	232	119	31	21	719	10
R5	3	8	60	197	41	23	300	295	25	16	968	6
R6	7	58	184	138	62	18	121	144	56	38	826	10
R7	2	14	117	94	59	20	173	79	91	23	672	10
R8	2	25	76	145	56	27	76	99	60	44	610	10
R9	4	10	80	146	148	27	150	83	70	45	763	10
R10	3	125	182	68	62	39	306	52	51	24	912	10
Total											8693	93

IV. RESULTS AND DISCUSSION

The test result is displayed in Table 1, showing the time each beginner participants spent completing each task (1 through 10) as well as the challenges encountered, which are color-coded with explanations provided in Table 2. The average time spent by respondents on tasks was 869.3 seconds. Task times for task 6 and task 10 were relatively shorter than for previous tasks, as tasks were performed sequentially, with task 6 conducted in the same menu as task 5 and task 10 in the same menu as task 9. Nevertheless, navigation difficulties persisted during these tasks.

TABLE 2. COLOR CODE AND CONSTRAINT DESCRIPTION

Color	Code	Constraint Description
Red	C1	Task not completed
Yellow	C2	Task completed but not using expected features
Blue	C3	Task completed but there are navigation difficulties
Orange	C4	Task completed but there was a technical error in the system
Purple	C2 + C3	Task completed but had difficulty navigating and did not use expected features

TABLE 3. NUMBER OF CONSTRAINT ON EACH TASK

Constraint	Tasks										Σ
	1	2	3	4	5	6	7	8	9	10	
C1	0	0	1	0	0	0	2	2	1	1	7
C2	0	0	3	1	0	0	5	0	0	0	9
C3	0	6	2	5	2	1	5	2	4	2	29
C4	0	0	0	1	0	0	0	0	0	0	1
Σ	0	6	6	7	2	1	12	4	5	3	46

The testing data was subsequently analyzed, yielding the number of issues encountered for each task, as shown in Table 3. Overall, tasks 2, 3, 4, and 7 were the most challenging for participants, highlighting several areas where the user experience could be improved. The primary issue identified, as illustrated in Table 3, was navigation difficulty (C3), which appeared 29 times throughout the responses. This problem became evident as many participants appeared feeling disoriented or confused while trying to follow the intended task flows. Some participants got lost within the interface or had

trouble identifying the correct steps, which led to a higher rate of incomplete tasks. Additionally, navigation errors were common, resulting in participants unintentionally activating unrelated features or landing on unintended pages. These recurring issues—task completion failures, navigation mistakes, and unintended feature use—suggest specific design elements in the interface that need refinement. Improving these aspects could significantly enhance the application's overall usability and provide a smoother experience for users.

A. Effectiveness

Table 4 presents the completion rate calculation results using (1) for the effectiveness of 10 respondents in completing 10 tasks, yielding an average ($\bar{E}$) of 93%. Mann-Whitney U-Test was conducted to verify the significance of the effectiveness difference between participants and the ideal performance of experts, who achieved a 100% completion rate, with $\alpha = 0.05$. The calculation using Python resulted in a U-statistic of 20 and a p-value of 0.3515, indicating no significant difference in effectiveness between beginner users and experts. This result suggests that, overall, participants were able to complete nearly all tasks with a very high success rate.

TABLE 4. COMPLETION RATE

Respondent Code	Completed Tasks	Respondent Completion Rate
R1	7	70%
R2	10	100%
R3	10	100%
R4	10	100%
R5	6	60%
R6	10	100%
R7	10	100%
R8	10	100%
R9	10	100%
R10	10	100%
Completion Rate		93%

However, task completion failures indicate that usability challenges remain within the design. To address these issues, the researchers utilized RTA data from each respondent who encountered difficulties in completing the tasks.

B. Efficiency

From the efficiency calculations using (2), with t_{oi} derived from the average optimal performance of expert users as shown in Table 5, a Process Efficiency ($\bar{P}_E$) of 45% was obtained. This indicates that, although respondents were able to complete their tasks and understand what needed to be done (with a relatively high completion rate of 93%), the average efficiency of beginner users was only 45% of expert efficiency. This finding suggests that the software design is not yet fully user-friendly for beginner users.

TABLE 5. OPTIMAL PERFORMANCE OF EXPERT USERS

Experts	Completion Time per Task (Seconds)										Total Time
	Task										
	1	2	3	4	5	6	7	8	9	10	
E1	4	10	63	61	47	25	70	81	35	18	414
E2	5	9	65	62	49	28	73	90	29	20	430
E3	5	8	69	61	50	19	68	85	32	17	414
E4	2	11	66	60	39	51	74	88	31	21	443
E5	3	8	60	65	41	23	70	91	25	16	402

C. Satisfaction

Satisfaction levels were derived from the UEQ analysis, represented by the Mean score, which is the average of all respondents' responses grouped according to each UEQ assessment scale.

TABLE 6. UEQ RESULT

Code	UEQ Scales	Mean	Comparison to benchmark
ATT	Attractiveness	0,350	Bad
PRS	Perspiciuity	0,000	Bad
EFF	Efficiency	0,575	Bad
DEP	Dependability	0,425	Bad
STI	Stimulation	0,550	Below Average
NOV	Novelty	-0,150	Bad

The satisfaction levels were then compared with the UEQ benchmark dataset, with results displayed in Table 6. The findings indicate that the scales for ATT, PRS, EFF, DEP, and NOV fall within the "Bad" category, or the bottom 25%. Meanwhile, STI, is categorized as "Below Average" compared to benchmarks from other products. The chart in Fig. 5 shows that user satisfaction across nearly all UX dimensions of the product falls within the low (Bad) category, with only one dimension (Stimulation) approaching the average, though still slightly below it.

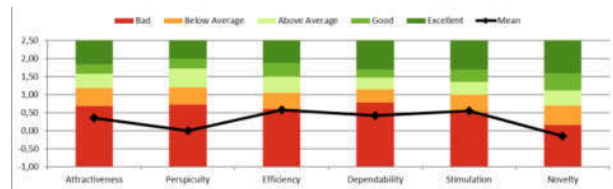


Fig. 5. UEQ Result Comparison with Benchmark

This indicates that users are generally dissatisfied with the product, particularly regarding visual appeal, ease of use, efficiency, and reliability. There is significant room for improvement in each of these dimensions to enhance the overall user experience and better meet user expectations.

D. Design Improvement Suggestion

During the RTA, respondents were asked to identify the obstacles they encountered. These obstacles were subsequently categorized based on their primary features, and potential improvements were analyzed. For instance, several respondents reported difficulties in discovering the status of their electronic certificates. R2 and R10 expressed similar sentiments, stating, "I'm confused about where the status is", while R4 remarked, "I just found out about the certificate status information in DS BRIN". This difficulty was categorized under the Digital Signature Status feature. The results of RTA analysis have been summarized into a design plan for improvements, as shown in Table 7.

TABLE 7. IMPROVEMENT SUGGESTIONS

Feature	Improvement Suggestion
Main menu	- Main menu reorganization. - Clarify the description on the menu and create clearer menu option - Displaying a summary of the number of unsigned documents along with available actions.
Digital Signature Status	Digital Signature status is displayed on the front page without having to select the menu
Single Document Signing	Integrate single document and multiple document features, as the only difference lies in the number of documents selected
Multiple Document Signing	- Refining the processes for adding files in the multiple file signing feature to accommodate cases where documents are in different folders - The functionality of the cross button in the document upload column of the Multiple File feature needs to be corrected; it should delete documents, not upload or send them. - Fix the button label for signing - Integrate single document and multiple document features, as the only difference lies in the number of documents selected
Signing Request	Integrate single document and multiple document features, as the only difference lies in the number of documents selected
Signing Approval	- Combine Single Document and Multiple Document Approval into one menu. - Make Passphrase Field popped up after User decided to accede or decline the document, not before - Removing the checklist button as it does not serve any functional purpose.

Based on the suggestions for improvement, a mapping was created to simplify the application features. The functionality for uploading documents and signing or requesting signatures from other users (whether for single or multiple documents) has been consolidated into a single feature. Similarly, the feature for signing approvals, which includes approvals for single documents, multiple documents, or documents for specific purposes, is also combined into one feature. Moreover, viewing signing history (whether for single or multiple documents) is also done through into a single feature. Fig. 6 showcases an improved layout of the DS BRIN main page, where the initial menu options have been reduced from 11 to 4 menus.

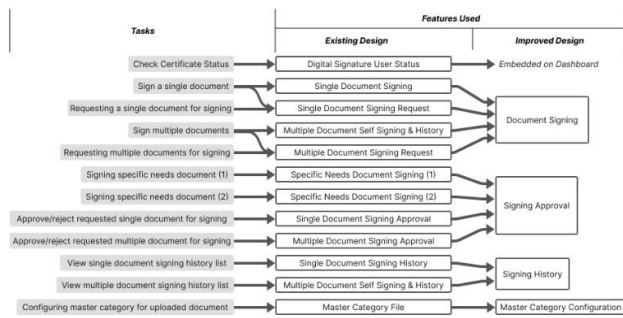


Fig. 6. Features Improvement Mapping

By minimizing menu choices, user hesitation and anxiety can be significantly decreased, thus enhancing the overall user experience [24] for example, Users no longer need to navigate to the "Digital Signature User Status" page to check the expiration status of their electronic certificates; this information is now available directly on the Dashboard, as shows in Fig. 7.

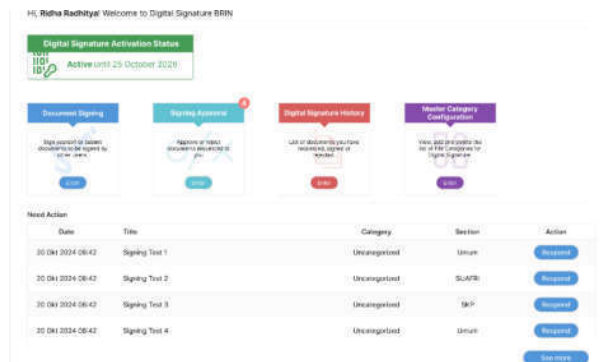


Fig. 7. DS BRIN Main Page Design Iteration

In the current version Multiple File Signing feature, if the User uploaded a document and then uploaded another, the prior document would be overwritten (deleted). Based on improvement suggestions, the button functionality has been updated: after the first file is uploaded, the button now changes to "Add More" and it should not overwrite the uploaded files. The document preview button has been enlarged and labeled to enhance visibility and the delete button has been renamed to "Remove All," with each file now having an individual delete button, as shows in Fig. 8.

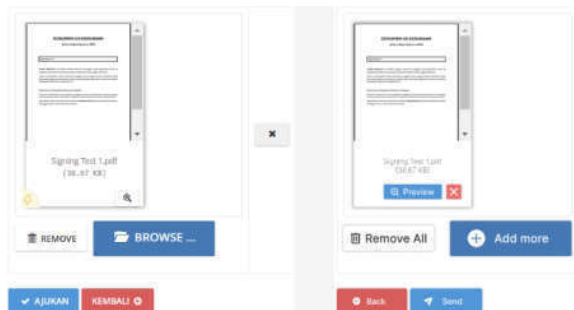


Fig. 8. Current Version of DS BRIN vs Design Improvement Suggestion

For the integrating signing feature, which merges submitting documents to other users with self-signing, an additional checklist button has been included, as shown in Fig. 9. This checkbox indicates that the user uploading the document is also a signer. If selected, the user will be directed to the signing module to proceed with the action. The title and field labels have also been changed from outside the field to inside it. When the user's name is entered, an "add more" inner field label appears, replacing the label from the previous design.

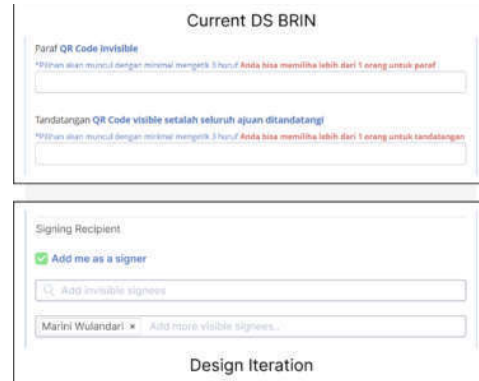


Fig. 9. Integrated Signer Recipient

Fig. 10 shows the integrated signing approval feature. In this design, a "Section" field and a filter button have been added to facilitate the integration of signing features for specific tasks, making it easier for users to sort documents according to their needs. The status column in the document list table has also been removed, as all documents on this page have a pending signature status, allowing more space to display file names for easier search and navigation.

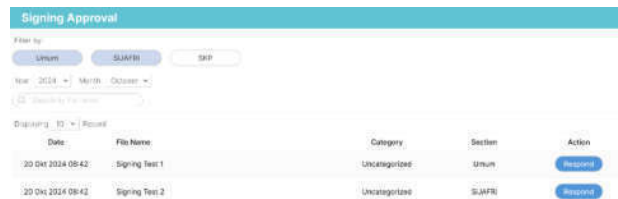


Fig. 10. Signing Approval

V. CONCLUSION

A. Summary

The usability analysis of the Digital Signature BRIN (DS BRIN) application indicates that the application is reasonably effective, with a completion rate of 93%, meaning that most users successfully completed the assigned tasks. Although the application has provided a link to the user manual document for using the application, users still have difficulty navigating, the application's efficiency remains low, with the average efficiency of respondents reaching only 45% of expert efficiency. This suggests that users, especially beginners, take longer to complete tasks due to navigation difficulties and non-intuitive feature usage. These issues are most pronounced in certain tasks, such as task 2, 3, 4, and 7, which frequently pose challenges and lead to users becoming lost or not using features

as intended. The most common issue faced by respondents was navigation difficulty, occurring in 29 tasks across all respondents. In terms of user satisfaction, results from the User Experience Questionnaire (UEQ) show that most UX aspects—including visual appeal, clarity, efficiency, and reliability—fall within the "Bad" category. Users reported low satisfaction with the application's design, particularly regarding ease of use and navigation. Based on the Retrospective Think Aloud findings, improvement recommendations can focus on reorganizing the main menu, simplifying navigation, and clarifying menu descriptions. Implementing these improvements is expected to enhance efficiency and user experience, making the application more user-friendly, particularly for beginners.

B. Further Research

Due to time and cost constraints, only 10 test users were able to participate in this study. With this limited sample size, we achieved results with an 80% probability that the true value differs by no more than 18% from the average measured value, which is adequate for practical development [21]. For future research, increasing the number of participants could enhance the confidence level of the results. To expand the usability analysis of digital signature applications, conducting in-depth evaluation methods, such as Eye-Tracking, can provide a more detailed understanding of how users navigate and interpret the application interface. Additionally, further studies could apply Importance-Performance Analysis (IPA) to assess which aspects of the application users consider important and evaluate the application's performance in meeting those expectations. Similar studies can also be conducted by comparing with similar application benchmarks to provide an idea of how this application should function according to similar application industry standards.

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